4.3 Speakers and Headphones

Briefly, speakers are devices that recreate a sound by converting electric signals into vibrations in the air with the use of an elastic diaphragm. The vibrations are translated into sound by the human ear. The human ear is capable of distinguishing vibrations ranging from 20 Hz to 20000 Hz. (1 Hertz (Hz) denotes one vibration per second).

4.3.1 Channels

A sound source can be recorded with many devices, and the stream recorded by each device is called a channel. Single-channel sound is called Mono, whereas two-channel sound (one each for the left and right sides) is called Stereo. Stereo playback creates a sense of space and direction due to the differences in the playback from each channel. A playback with five or more channels (4.1) is called multichannel.

4.3.3 Frequency Response

The range of frequencies that can be reproduced by the speaker is its frequency response. In actuality, no single speaker can cover the entire audible range of the human ear.

4.3.4 Multiway Speakers

A multiway speaker contains more than one type of driver in the same enclosure. Depending on the number of types of drivers in a speaker, the speaker can be classified as 2-way (2 drivers in the same enclosure), 3-way, etc. The different driver types—tweeter, midrange, woofer, full range—cover the entire audio spectrum.

4.3.5 PMPO

Peak Music Power Output is the maximum output capacity of a speaker. There is no standard procedure to calculate PMPO, and since this value cannot be sustained by the speaker for more than a brief period of time, it is not a reliable measure of the speaker's capabilities.